

Midterm Exam
Logic and Sets for AI

*This exam has **3 pages** and **7 exercises** with a total of **80 points**. The result will be computed as $1 + \left(\frac{\text{total}}{80} \cdot 9\right)$*

Note that in this course, 0 is considered a natural number. If you are, in any exercise, unsure about the definition of some term/operator/... state your assumed definition at the beginning of the exercise.

*Please put **your name on every sheet** you hand in, and write as clearly as possible. Submissions (even parts) without a name might **not be considered**. If we cannot read a statement it will be considered **incorrect**.*

Good luck!

1. Semantic entailment of Sheffer Stroke (10 points)

Determine if the following semantic entailment holds:

$$(p \rightarrow (q \oplus r)) \wedge ((q \mid \neg r) \vee p) \models \neg q \leftrightarrow r$$

Solution: In view of the fifth and eighth row in the truth table below, this semantic entailment does not hold.

p	q	r	$q \oplus r$	$p \rightarrow (q \oplus r)$	$\neg r$	$q \mid \neg r$	$(q \mid \neg r) \vee p$	$(p \rightarrow (q \oplus r)) \wedge ((q \mid \neg r) \vee p)$	$\neg q$	$\neg q \leftrightarrow r$
T	T	T	F	F	F	T	T	F	F	F
T	T	F	T	T	T	F	T	T	F	T
T	F	T	T	T	F	T	T	T	T	T
T	F	F	F	F	T	T	T	F	T	F
F	T	T	F	T	F	T	T	T	F	F
F	T	F	T	T	T	F	F	F	F	T
F	F	T	T	T	F	T	T	T	T	T
F	F	F	F	T	T	T	T	T	T	F

p	q	r	$p \vee r$	$p \wedge q$	$\neg(p \wedge q)$	$\neg(p \wedge q) \rightarrow r$	$(p \vee r) \vee (\neg(p \wedge q) \rightarrow r)$	$(p \vee \neg q \vee r) \wedge (p \vee q \vee r)$
T	T	T	T	T	F	T	T	T
T	T	F	T	T	F	T	T	T
T	F	T	T	F	T	T	T	T
T	F	F	T	F	T	F	T	T
F	T	T	T	F	T	T	T	T
F	T	F	F	F	T	F	F	F
F	F	T	T	F	T	T	T	T
F	F	F	F	F	T	F	F	F

2. CNF (4 + 4 + 2 points)

- (a) Construct a formula in CNF that is semantically equivalent to a formula

$$(p \vee r) \vee (\neg(p \wedge q) \rightarrow r)$$

using **the truth table method**.

Solution: From the truth table of $(p \vee r) \vee (\neg(p \wedge q) \rightarrow r)$ we can observe that in the rows 6 and 8 the formula is false. Rows 6 and 8 are therefore the only relevant ones for constructing the CNF.

In row 6 we have the following variable assignments, $p = F$, $q = T$ and $r = F$. Therefore, the CNF clause for this row is $(p \vee \neg q \vee r)$. In row 8, $p = F$, $q = F$ and $r = F$. This gives the CNF clause: $(p \vee q \vee r)$. The CNF for the formula $(p \vee r) \vee (\neg(p \wedge q) \rightarrow r)$ is:

$$(p \vee \neg q \vee r) \wedge (p \vee q \vee r)$$

- (b) Determine a semantically equivalent CNF of the formula

$$\neg((p \vee q) \rightarrow r) \wedge (\neg p \rightarrow r)$$

by **applying the algorithm CNF**.

Solution: Applying the CNF algorithm to the formula $\neg((p \vee q) \rightarrow r) \wedge (\neg p \rightarrow r)$.

- (i) Implication free: $\neg(\neg(p \vee q) \vee r) \wedge (\neg\neg p \vee r)$
- (ii) NNF: $(\neg\neg(p \vee q) \wedge \neg r) \wedge (p \vee r)$
- (iii) NNF: $((p \vee q) \wedge \neg r) \wedge (p \vee r)$

$$(p \vee q) \wedge \neg r \wedge (p \vee r)$$

- (c) Explain and apply the criterion for determining whether a CNF is a tautology to the CNF you constructed in (b).

Solution: A CNF is a conjunction of one or more clauses, where each clause is a disjunction of literals. A conjunction can only be a tautology if all its conjuncts are tautologies. So each clause in a CNF must be a tautology. The clauses of a CNF are disjunctions which can only be a tautology, if and only if they contain literals p and $\neg p$ for some p . None of the clauses of the CNF from b meet this condition so the CNF can not be a tautology.

3. DPLL (9 points)

Apply the DPLL procedure to the CNF

$$(p \vee \neg r) \wedge (q \vee \neg r) \wedge (\neg p \vee \neg q) \wedge (q \vee r)$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Unless there is unit clause or a pure literal, let the procedure branch **first on p, then on q, and then on r**; and let it always try the truth value true ($\top$) before false ($\perp$).

If the formula is satisfiable, please provide a model.

Solution: There is no unit clause or pure literal, so we set p to be true, which produces the formula

$$(q \vee \neg r) \wedge \neg q \wedge (q \vee r)$$

Now $\neg q$ is a unit clause, which means q has to be false. This produces the formula

$$\neg r \wedge r$$

which is unsatisfiable. Thus we need to backtrack on p and set it to false, which results in the following formula:

$$\neg r \wedge (q \vee \neg r) \wedge (q \vee r)$$

$\neg r$ is a unit clause, so r has to be false, resulting in the formula:

$$q$$

q is a unit clause, so q has to be true.

The formula is thus satisfiable, with a model $p = \perp, q = \top, r = \perp$

t_X	t_Y	t_Z	$t_Y \iff t_Z$	$t_X \iff \neg t_Y$	$t_Y \iff (t_Y \iff t_Z)$	$t_Z \iff t_X$
T	T	T	T	F	T	T
T	T	F	F	F	F	F
T	F	T	F	T	T	T
T	F	F	T	T	F	F
F	T	T	T	T	T	F
F	T	F	F	T	F	T
F	F	T	F	F	T	F
F	F	F	T	F	F	T

4. Island Puzzle (6 + 5 points)

- (a) On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders Xander, Yara and Zain.

Xander says: “Yara is a liar”

Yara says: “I am the same as Zain”

Zain says: “Xander is a truth speaker”

Determine by means of logical reasoning or by use of a truth table whether islanders X, Y and Z are truth speakers or liars.

Solution: If it is assumed that X is truthful, that must mean Y is a liar, which implies that Z is truthful, as is confirmed by his statement. So X truthful, Y liar and Z truthful holds. On the other side, if we assume X is a liar, that would mean Y is truthful, which would mean Z is truthful as well. Since Z says that X is truthful but we assumed X was lying, this does not hold and we encounter a paradox. This means that only the first combination works.

Alternatively, looking at the truth table shows that the only situation which holds is the one in which the three statements combined with truth statements on the agents hold. This is T, F, T for X, Y, Z.

NOTE: $t_Y \iff t_Z$ in Yara’s statement is interchangeable with $(t_Y \wedge t_Z) \vee (\neg t_Y \wedge \neg t_Z)$

- (b) On this same island, you are now facing two different islanders, Abraham and Bethel knowing that one of them is a truth speaker and the other is a liar but not knowing who is who. A road bifurcates in front of you, one path leading to a treasure and the other leading to peril. Both islanders know each other’s nature and the characteristics of the paths. With one clever question yes or no question that you might ask any one of them, you can be sure to find the safe path. What is the question and what is the direction which you would then take?

Solution: A question which neutralizes the lack of knowledge is needed, so something that both islanders would give the same answer to. Such a question could be ‘Would the other islander tell me to go this way?’. One should then go in the opposite direction. If the liar answered, in fact, they would lie about the correct answer of the other. If the truth speaker answered, they would be honest about the incorrect answer that the other would give.

5. Sets (4 + 9 points)

Consider the following set-theoretic equality:

$$(A \cap B) \cap (A \Delta B) = \emptyset \quad (1)$$

- (a) Prove this identity using a sequence of Venn diagrams. You can draw a series of diagrams showing how one side of the equality is constructed from A and B via elementary set operations and use this to prove that the equation holds for any two sets A and B . Please label the diagrams so that it is clear which elementary operation or formula they correspond to.

Solution: The solution is shown in Figure 1.

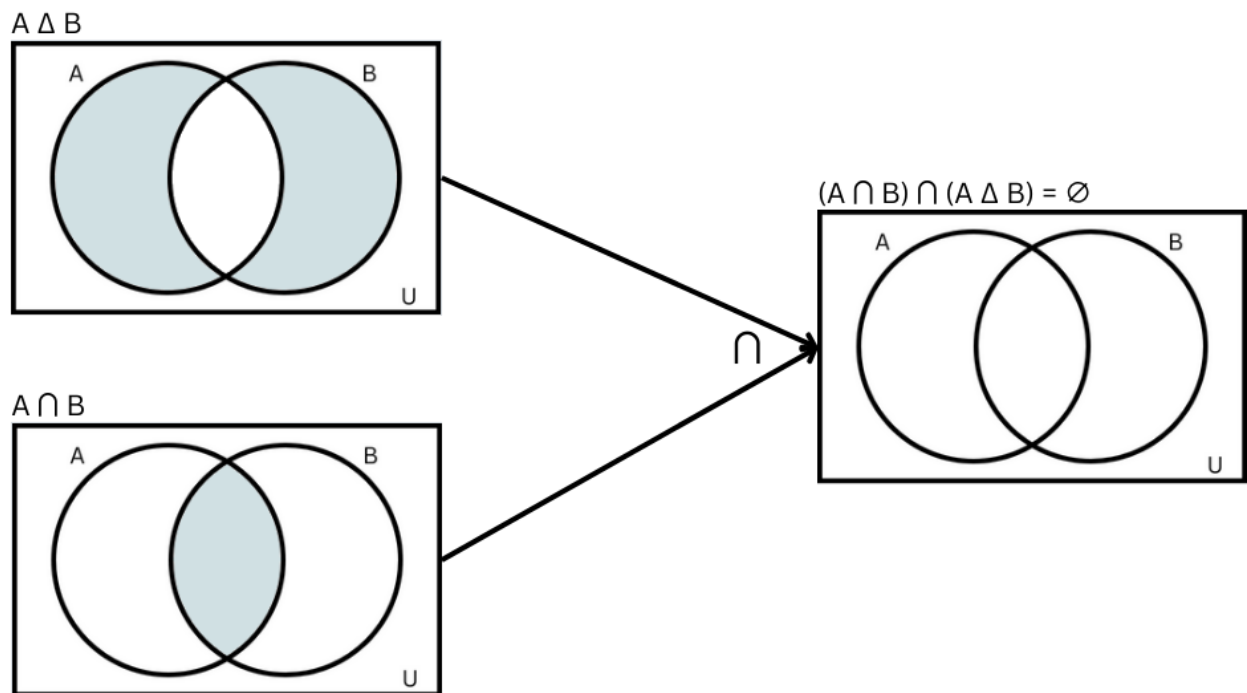


Figure 1: Solution to Exercise 5.(a)

- (b) Prove the identity using the algebra of sets. Please make sure to name all the rules which you apply. You may name more than one rule per line if it is clear and correct how you apply them.

Solution: Steps of the derivation:

$$(A \cap B) \cap (A \Delta B)$$

definition of Δ : $(A \cap B) \cap ((A \setminus B) \cup (B \setminus A))$

distributivity: $((A \cap B) \cap (A \setminus B)) \cup ((A \cap B) \cap (B \setminus A))$

definition of $\setminus$: $((A \cap B) \cap (A \cap B')) \cup ((A \cap B) \cap (B \cap A'))$

associativity and commutativity: $(A \cap A \cap B \cap B') \cup (A \cap A' \cap B \cap B)$

idempotence and complement: $(A \cap \emptyset) \cup (\emptyset \cap B)$

domination: $\emptyset \cup \emptyset$

idempotence: $\emptyset$

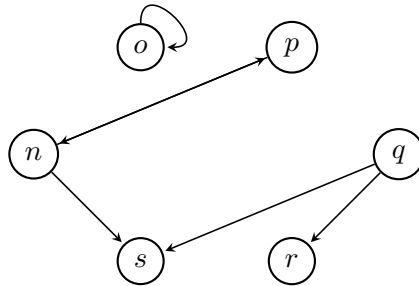
6. Relations (4 + 6 + 6 points)

Given the set $V = \{n, o, p, q, r, s\}$ with the relations

$$A = \{\langle n, p \rangle, \langle n, s \rangle, \langle o, o \rangle, \langle p, n \rangle, \langle q, r \rangle, \langle q, s \rangle\}$$

$$B = \{\langle n, n \rangle, \langle n, q \rangle, \langle q, r \rangle\}$$

- (a) Given the relation A, draw the directed graph representation.



Solution:

- (b) Is the relation A reflexive? Is A transitive? Is B anti-symmetric? Why or why not?

Solution: The definition of reflexivity states that for all x, we must have the relation xRx . We thus conclude that A is NOT reflexive, because not ALL elements relate to themselves. There are missing:

$$missing = \{\langle n, n \rangle, \langle p, p \rangle, \langle q, q \rangle, \langle r, r \rangle, \langle s, s \rangle\}$$

The definition of transitivity states that $\forall x, y, z : xRy \wedge yRz \rightarrow xRz$. We then conclude that A is not transitive, because we have $\langle p, n \rangle, \langle n, s \rangle$, but not $\langle p, s \rangle$; also we have $\langle p, n \rangle, \langle n, p \rangle$, but not $\langle n, n \rangle$ and not $\langle p, p \rangle$.

The definition of anti-symmetry states that $\forall x \neq y : xRy \rightarrow \neg yRx$. So yes, B is anti-symmetric.

- (c) Draw a Venn diagram depicting the composition $A^{-1} \circ B$ and write down the matrix representation of it.

Solution: The Venn Diagram representation is shown in Fig.2.

The matrix representation is:

$$A^{-1} \circ B = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (2)$$

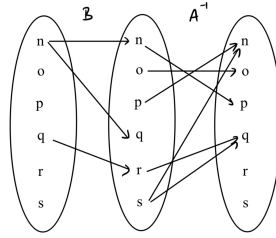


Figure 2: Solution to Exercise 6.(c)

7. Ordering relations (7 + 4 points)

Let P be a subset of $\{\text{Morning, Afternoon, Evening, Night}\}^2$ and be defined as (with elements abbreviated by their first letters):

$$P := \{\langle M, M \rangle, \langle M, A \rangle, \langle A, M \rangle, \langle A, A \rangle, \langle E, E \rangle, \langle E, N \rangle, \langle N, E \rangle\}$$

Consider the Cartesian ordering relation on P which is induced by the order in which the times of day follow each other, i.e., $M \leq A \leq E \leq N$.

- (a) Construct and draw the Hasse diagram of this Cartesian order on P . You can use the algorithm which you learnt about in class, but you are not required to do so. For the sake of simplicity, you can abbreviate pairs such as $\langle M, A \rangle$ by writing MA .

Solution: The solution is shown in Figure 3.

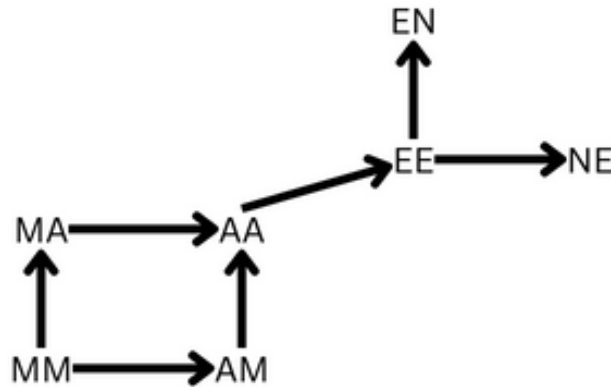


Figure 3: Solution to Exercise 7.(a)

- (b) Determine and list all the minimal and maximal elements in P according to this Cartesian ordering. Is there a minimum and/or maximum? If yes, what are these?

Solution: Minimal element is MM , and it is also a minimum. Maximal elements are EN and NE , but there is not a maximum.

End of the exam.